



DocForge - Document Forgery Detection System

Prasad Jadhav	23107097
Dhananjay Agarwal	23107121
Shardul Kadam	23107068
Riddhi Manchekar	23107109

Project Guide
Mrs. Hardiki Achrekar

Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement
- System Design
- Technologies and methodologies
- Implementation
- Conclusion
- References

Sustainable Development Goals (SDG) mapped

- ▶ **SDG 4 – Quality Education**
- ▶ Detects forged academic certificates and transcript0



Introduction

- The use of **digital documents** such as bank statements, academic certificates, identity proofs, and legal agreements has increased rapidly.
- This growth has also led to a **rise in document forgery and manipulation**.
- Forged documents are commonly used for **financial fraud, fake job applications, loan approvals, and identity theft**.
- Current verification methods mainly rely on **manual inspection**, where officials visually check documents for inconsistencies.
- **Manual verification is time-consuming, error-prone, and inefficient** when handling large numbers of documents.
- Modern editing tools can create **highly realistic document alterations**, making manual detection difficult.
- Therefore, an **automated system for document forgery detection** is required.
- This project proposes a **Data Forgery Detection System** using **image preprocessing, Optical Character Recognition (OCR), and machine learning algorithms**.
- The system aims to **detect tampered regions, identify textual inconsistencies, and classify documents as genuine or forged**, improving the **accuracy, speed, and reliability** of document verification.

Introduction

Objectives:

- To preprocess document images using grayscale conversion and enhancement to highlight forgery artifacts.
- To extract textual content using Optical Character Recognition (OCR) and analyze inconsistencies for tampering detection.
- To classify documents as forged or genuine using Support Vector Machine and Random Forest algorithms.
- To evaluate system performance using accuracy, precision, recall, and F1-score metrics.

Features:

- Automated document preprocessing to enhance tampered regions.
- OCR-based text extraction for analyzing textual authenticity.
- Machine learning-based classification for forgery detection.
- Dual analysis approach combining visual artifact detection and textual verification.

Literature Survey of the existing system

Sr. No.	Title (IEEE format)	Authors	Year	Outcomes	Methodologies / Technologies used	Demerits / Research gap
1	“Find It Again: A Dataset and Baselines for Receipt Document Forgery Detection”	B. M. Tornés, et al.	2023	Released an open receipt forgery dataset and baseline results for both image-level and text-aware forgery detection; baselines show visual + text features needed.	Public dataset; image-based baselines (CNN) and text-aware baselines (OCR + text matching); evaluation metrics: precision/recall/F1.	Dataset limited to receipts; visual baselines detect texture anomalies but semantic text inconsistencies are still hard.
2	“Document Forgery Detection” (IJEAT open PDF)	N. Nandini, K. Keerthi Joshi, M. Devprakash, et al.	2023	Demonstrates a system detecting signature and copy-move forgeries using Error Level Analysis (ELA) + deep features; reports strong accuracy on their dataset.	Preprocessing (ELA), feature extraction (CNN / capsule layers), similarity measures for copy-move detection, evaluation on custom dataset.	Uses image forensic cues only (ELA, capsule nets); limited generalization to printed/scanned text edits; lacks OCR semantic validation.

Literature Survey of the existing system

Sr. No.	Title (IEEE format)	Authors	Year	Outcomes	Methodologies / Technologies used	Demerits / Research gap
3	“Document Forgery Detection using Machine Learning” (IJRPR open PDF)	(Authors: Nandini et al. / IJRPR)	2024	Presents capsule networks and ELA plus ML classifiers to detect forgeries; good signature forgery detection on lab datasets.	Capsule networks, ELA, classical ML classifiers and CNN feature extractors.	Focus on signatures and copy-move; not evaluated on large, heterogeneous scanned document corpora; no semantic text anomaly model.
4	“Enhancing Document Forgery Detection with Edge-Focused Deep Learning”	Y. Y. Bae	2025 (open access)	Introduces edge-attention modules that improve detection of subtle tampering near character/edge boundaries; outperforms baseline CNNs on benchmark manipulations.	CNN backbones (DenseNet / ResNet) + edge-attention modules, Sobel/pre-edge concatenation, multi-scale supervision.	Strong visual edge detection but no textual semantic consistency checks; vulnerable to edits that change text semantics without strong edge artifacts.

Literature Survey of the existing system

Sr. No.	Title (IEEE format)	Authors	Year	Outcomes	Methodologies / Technologies used	Demerits / Research gap
5	“AI-Based Document Forgery Detection Using OCR and Convolutional Neural Networks”	M. Bhatia, N. Mehta	2022	Proposes a hybrid pipeline combining OCR preprocessing with CNN-based classification for detecting forged document regions; achieved ~93.8% accuracy on synthetic dataset.	Tesseract OCR, OpenCV preprocessing, CNN architecture with convolution + max-pooling layers, TensorFlow/Keras training, evaluation metrics (accuracy, precision, recall).	Dataset size small (~1200 synthetic documents); limited diversity of document types; performance affected by low-quality scans and uneven backgrounds.
6	“FantasyID: A Dataset for Detecting Digital Manipulations of ID- Documents”	P. Korshunov, A. Mohammadi, Vidit, C. Ecabert, S. Marcel	2025	Introduces FantasyID dataset containing realistic ID cards with multiple languages and manipulation types to benchmark forgery detection algorithms in KYC systems.	Dataset generation with synthetic ID templates, face-swap and text-inpainting manipulations, evaluation with modern forgery detection models (TruFor, MMFusion, UniFD, FatFormer).	Detection algorithms show high false-negative rates (~50%) in realistic conditions; highlights difficulty of detecting sophisticated digital manipulation in ID verification pipelines.

Literature Survey of the existing system

Sr. No.	Title (IEEE format)	Authors	Year	Outcomes	Methodologies / Technologies used	Demerits / Research gap
7	“MIDV-2020: A Comprehensive Benchmark Dataset for Identity Document Analysis”	K. B. Bulatov, E. V. Emelianova, D. V. Tropin, N. S. Skoryukina, Y. S. Chernyshova, A. V. Sheshkus, S. A. Usilin, Z. Ming, J.-C. Burie, M. M. Luqman, V. V. Arlazarov	2022	Presents a large dataset of identity documents with annotated images, scans, and videos to support research in document recognition and fraud detection.	Document detection, segmentation, text recognition (Tesseract), face detection (CNN models), evaluation of document location algorithms and segmentation methods.	Limited diversity of real document samples; many images derived from same templates; primarily focused on recognition tasks rather than direct forgery detection.

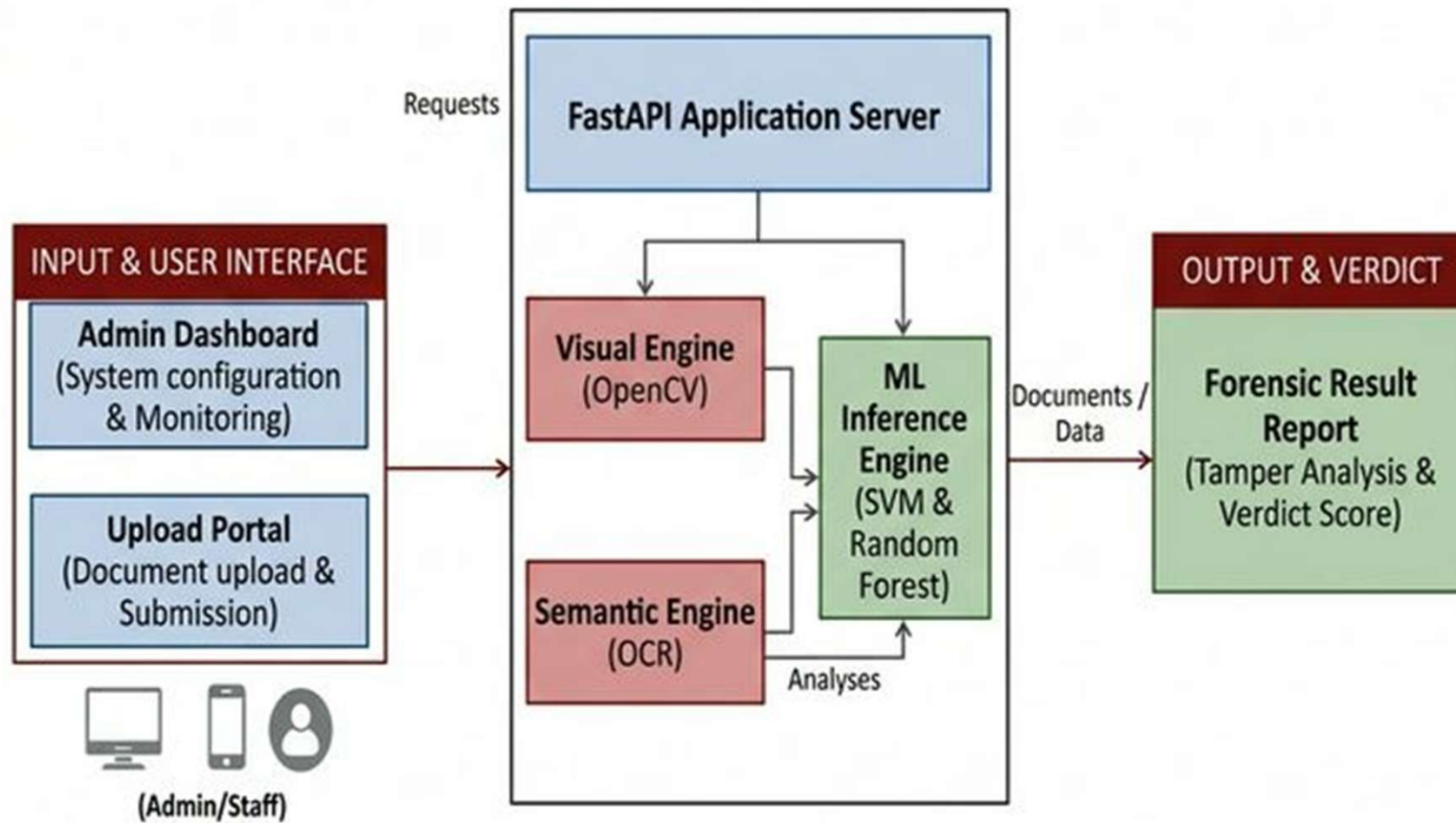
Limitations of existing systems

- Difficulty Detecting Semantic Text Manipulation - Most visual-based forgery detection methods identify texture or pixel anomalies but struggle to detect semantic inconsistencies in textual information.
- Limited Detection of Subtle Digital Edits - Some sophisticated manipulations such as text replacement without strong edge artifacts or texture changes may not be easily detected.
- Dataset Size Constraints - Training datasets may contain synthetic or limited samples, which can affect the model's performance when applied to diverse real-world documents.
- Focus on Specific Forgery Types - Many approaches primarily detect signature forgery or copy-move attacks, while other manipulation types may remain challenging.

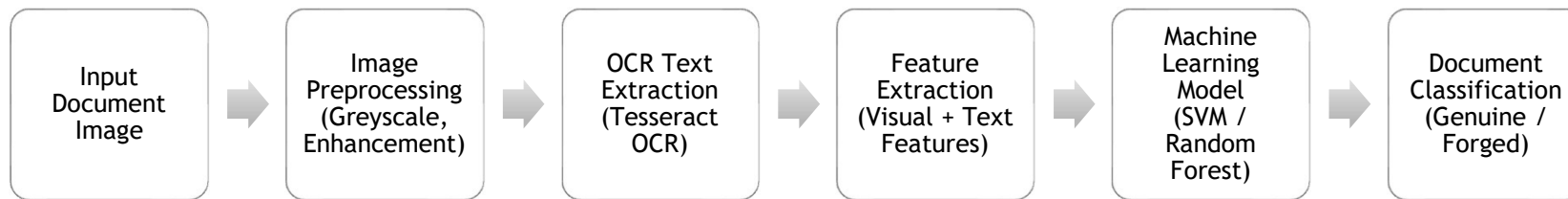
Problem statement

- Digital documents such as bank statements, academic certificates, identity cards, and legal agreements are widely used for verification.
- With the availability of advanced editing tools (Photoshop, scanners, document editors), creating forged or manipulated documents has become easier.
- Fraudulent documents are often used for loan approvals, fake job applications, financial fraud, and identity theft.
- Current verification methods rely mainly on manual inspection, which is time-consuming, error-prone, and inefficient when processing large numbers of documents.
- Therefore, there is a need for an automated system that can detect document forgery by analyzing visual artifacts and textual inconsistencies using machine learning and OCR techniques.

System Architecture



System Design



Components

- Image Preprocessing: Enhances document image to highlight tampering artifacts.
- OCR Extraction: Converts document text into machine-readable format.
- Feature Extraction: Detects inconsistencies in visual patterns and text structure.
- Machine Learning Model: Uses SVM and Random Forest to classify documents.
- Output: Identifies whether the document is genuine or forged.

Technologies and methodologies

For Image Preprocessing:

- Grayscale Conversion
- Noise Removal
- Contrast Enhancement

For Text Extraction:

- OCR (Tesseract Engine)

For Forgery Classification:

- Support Vector Machine (SVM)
- Random Forest Classifier

Implementation

Document Forgery Detection System

Document Forgery Detection

Advanced AI-powered system to detect forged documents using OCR analysis and deep learning models with high accuracy, precision, and recall.

Upload Document



Drag & drop a document here, or click to select

Supports: PDF, PNG, JPG, JPEG

Analysis Results



Upload a document to see analysis results

Implementation

Document Forgery Detection

Advanced AI-powered system to detect forged documents using OCR analysis and deep learning models with high accuracy, precision, and recall.

Upload Document



dj_adhar.jpeg
0.08 MB



 Analyze Document

Analysis Results



Document is Authentic

Confidence: 50.78%

Processing Parameters

Preprocessing: Applied

Grayscale: Enabled

Enhancement: CLAHE / Contrast
Boost

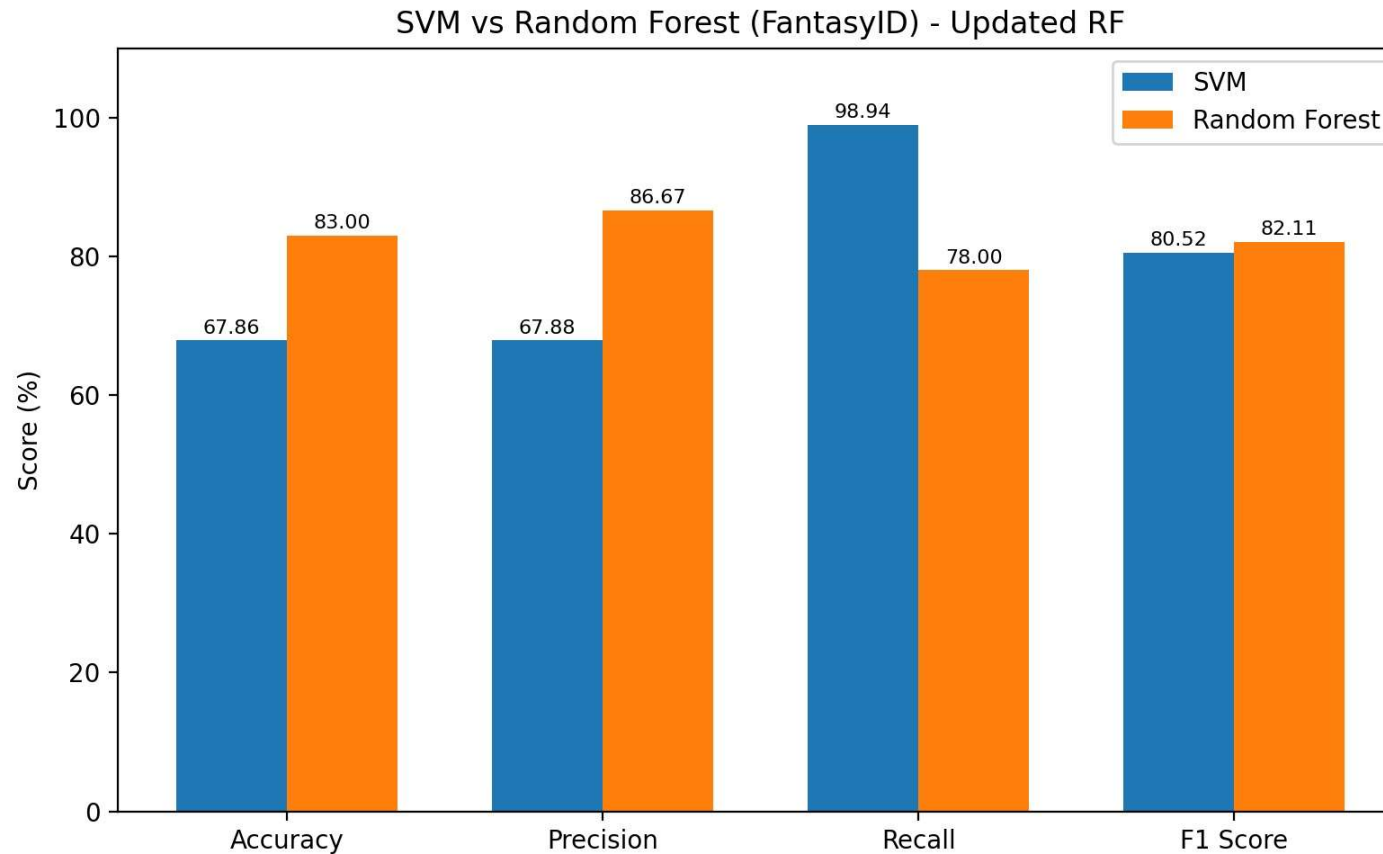
OCR: Extracted (67 chars)

OCR Extracted Text

Dhananjay Agarwal
DOB : 13/09/2005
Male a

o

Result



- **Random Forest** has higher **accuracy** and **precision**, so it's more reliable overall.
 - **SVM** has very high **recall**, so it catches more actual positives but makes more mistakes.
 - **F1 score** is slightly better for Random Forest, meaning better balance.
- Random Forest is better overall, SVM only if recall matters most.

Conclusion

- Automatically detects manipulated or tampered documents
- Analyzes both visual artifacts and textual inconsistencies
- Uses image preprocessing and OCR for enhanced analysis
- Applies SVM and Random Forest for accurate classification
- Extracts features to identify suspicious regions
- Reduces human effort, improves accuracy, and scales efficiently

References

- [1] B. M. Tornés, J. P. N. Pardo, and D. Karatzas, “Find It Again: A Dataset and Baselines for Receipt Document Forgery Detection,” in Proceedings of the International Conference on Document Analysis and Recognition (ICDAR), 2023.
- [2] N. Nandini, K. Keerthi Joshi, M. Devprakash, and A. Kumar, “Document Forgery Detection,” International Journal of Engineering and Advanced Technology (IJEAT), vol. 12, no. 3, pp. 45–50, 2023.
- [3] N. Nandini, K. Keerthi Joshi, and M. Devprakash, “Document Forgery Detection using Machine Learning,” International Journal of Research Publication and Reviews (IJRPR), vol. 5, no. 2, pp. 1120–1125, 2024.
- [4] Y. Y. Bae, “Enhancing Document Forgery Detection with Edge-Focused Deep Learning,” IEEE Access, vol. 13, pp. 1–12, 2025.
- [5] M. Bhatia and N. Mehta, “AI-Based Document Forgery Detection Using OCR and Convolutional Neural Networks,” International Journal of Artificial Intelligence and Applications, vol. 13, no. 4, pp. 85–92, 2022.
- [6] P. Korshunov, A. Mohammadi, Vidit, C. Ecabert, and S. Marcel, “FantasyID: A Dataset for Detecting Digital Manipulations of ID-Documents,” arXiv preprint arXiv:2507.20808, 2025.
- [7] K. B. Bulatov, E. V. Emelianova, D. V. Tropin, N. S. Skoryukina, Y. S. Chernyshova, A. V. Sheshkus, S. A. Usilin, Z. Ming, J.-C. Burie, M. M. Luqman, and V. V. Arlazarov, “MIDV-2020: A Comprehensive Benchmark Dataset for Identity Document Analysis,” Computer Optics, vol. 46, no. 2, pp. 252–270, 2022.

Thank You...!!